

Methodology Summary: Ranking U.S. States by Opportunity for Single Mothers

1. Overview

This document outlines the methodology used to rank all 50 U.S. states based on their suitability for single mothers. The ranking is derived from a composite scoring model that synthesizes key metrics across affordability, income, employment, healthcare, education, and safety. The goal is to provide a data-driven, transparent, and reproducible assessment to support decision-making.

2. Data Sources and Metrics

Data was extracted from authoritative public sources for a consistent reference period (primarily 2023-2024). The following metrics were used:

Category	Metric	Source	Year(s)	Notes
Affordability	Housing Cost Burden (renter >30%)	ACS 1-Year (DP04)	2024	Percent of renter households with high cost burden.
	Childcare Costs (composite)	CCAoA	2024	Weighted avg. of infant, toddler, and preschool center-based care.
	Food & Goods Costs (RPP)	BEA RPP	2023	Regional Price Parity for goods, relative to U.S. average.
	Transportation Burden (commute)	ACS 1-Year (S0801)	2024	Mean travel time to work in minutes.
Income & Employment	Median Income (single mother)	ACS 1-Year (S1903)	2024	Median income for female householder, no spouse present.
	Unemployment Rate	BLS LAUS	2024	Annual average of seasonally adjusted state unemployment rate.

Category	Metric	Source	Year(s)	Notes
Healthcare	Uninsured Rate (female)	ACS 1-Year (S2701)	2024	Percent of female population without health insurance.
Education	High School Graduation Rate (ACGR)	NCES	2020	Adjusted Cohort Graduation Rate (most recent complete).
Safety	Crime Rates (violent & property)	FBI SAPI	2022/23	<i>Pending API key enablement. Placeholder used in current model.</i>

Data Portals & APIs Used:

- U.S. Census Bureau API: `api.census.gov`
- Bureau of Labor Statistics API: `api.bls.gov`
- Bureau of Economic Analysis API: `apps.bea.gov/api`
- Child Care Aware of America (CCAoA): `childcareaware.org`
- National Center for Education Statistics (NCES): `nces.ed.gov`
- FBI Crime Data Explorer API: `api.usa.gov/crime/fbi/sapi`

3. Data Processing and Standardization

- Extraction:** Data was programmatically pulled from APIs where available. Manual downloads were used for sources without stable public APIs (e.g., CCAoA, NCES).
- Cleaning:** Each dataset was cleaned to ensure consistent formatting. State identifiers were standardized to a zero-padded two-digit FIPS code. Numeric columns were coerced to appropriate data types.
- Harmonization:** All datasets were filtered to include only the 50 U.S. states, ensuring a consistent scope for ranking.
- Merging:** The cleaned datasets were merged into a single master table using the state FIPS code as the primary key.

4. Composite Scoring Model

a. Normalization

To ensure comparability across metrics with different units and scales, each metric was normalized to a 0-100 score. The process involved:

- Winsorization:** To reduce the effect of extreme outliers, values were capped at the 5th and 95th percentiles.
- Min-Max Scaling:** The winsorized values were then scaled to a 0-100 range. For metrics where a lower value is better (e.g., costs, unemployment), the formula was inverted:
 - **Higher is better:** `Score = 100 * (Value - Min) / (Max - Min)`
 - **Lower is better:** `Score = 100 * (Max - Value) / (Max - Min)`

b. Weighting

A weighted-average approach was used to calculate the final composite score. The weights were assigned based on their relative importance to the overall opportunity for a single mother. The weights for the available metrics were renormalized to sum to 100%.

Category	Metric	Base Weight (%)
Affordability	Housing, Childcare, Food, Transport	35
Income & Employment	Median Income, Unemployment	25
Healthcare	Uninsured Rate	10
Education	Graduation Rate (ACGR)	15
Safety	Crime Rates	(15) - Pending

c. Final Score Calculation

The final composite score for each state was calculated as the weighted sum of its normalized scores:

$$\text{Composite Score} = \sum (\text{Normalized Score}_i * \text{Weight}_i)$$

States were then ranked from 1 (best) to 50 (worst) based on their final composite score.

5. Validation

- **Outlier Analysis:** States at the extremes of each normalized metric were reviewed to ensure they were plausible.
- **Sensitivity Analysis:** The rankings were tested for robustness by running two alternative scenarios:
 1. **Affordability-Focused:** Increased the weight of all affordability metrics to 50%.
 2. **Z-Score Normalization:** Replaced min-max scaling with z-score normalization.
- The rank changes were analyzed to confirm that the model was not overly sensitive to minor changes in weighting or normalization methodology.

This document provides a summary of the methodology. For detailed data, please refer to the accompanying Excel workbook.